

Top-down decoy feature detection — score distributions, ROC/AUC, operating thresholds

ProteoStar untargeted MS1 feature detector — branch `decoy-features` 14 July 2026

Replicates the bottom-up envelope-decoy experiment (`analysis/decoy_features/`) on the two **top-down** datasets. For each dataset we detect features four ways — the real averagine target plus three decoy envelopes — score every feature with a 2-D and 3-D envelope-fit cosine against the model it was detected with, and ask how well the score separates real (ID-matched) features from decoys.

The four envelope models

A decoy replaces the averagine isotope envelope — used by the detector’s matched filter, its refinement, and the score, end to end — with a theoretical envelope no real proteoform can produce, so any feature it finds is found on coincidence. The three decoys are:

- **shifted spacing** — the real averagine tooth *weights*, but teeth laid at 0.94 Da instead of the 13C 1.00336 Da (an off-lattice comb).
- **weird averagine (Fe+Cl)** — an exotic per-unit composition (P2 C1 N1 Cl1 Fe1) that *replaces* the peptide backbone; the 37Cl / 57,58Fe A+2 satellites push the mode far above the monoisotope, a tooth-weight profile no peptide makes.
- **shuffled envelope** — the real averagine tooth weights randomly permuted (on-lattice, correct multiset, wrong order).

Figure 1 shows all four at a top-down-relevant 12,000 Da, each generated so its **most-abundant peak sits at the same mass (12,000 Da) and is normalized to 1** — only the envelope shape differs.

Key enabling change: decoys threaded through the top-down detector

The bottom-up decoy work threaded the decoy envelope + lattice through the **serial single-charge** path only; the **top-down joint charge-ladder detector** (`detect_features_multicharge` → `score_mass_hypothesis` / `refine_mono_offset` / `gather_charge_extent` / `emit_mass_hypothesis` in `trace_kernel.rs`) was hardcoded to `averagine_intensities_from_mono` at physical C13/z spacing, so `COMB_MODEL` / `DECOY_LATTICE` had no effect on it. Those functions (and the seed spacing screen) now use model-aware weights (`multicharge_weights`) and `tooth_offsets` / `tooth_step` under the configured `LatticeMode`. The averagine / Uniform target path is **bitwise unchanged** (Averagine maps back to the same call, Uniform back to `k.C13/z`); only the decoy models change behaviour. This is what makes decoy “features” — found on coincidence by an envelope no real proteoform makes — appear at all in top-down.

Configuration (identical across all runs — envelope model is the only variable)

`TOPDOWN=1 APEX_PREGATE=0 REFINE_METHOD=shift_apex TD_MONO_FIT=0 ISODEC_CHARGE=0`

`shift_apex` refine is decoy-aware (Deconvoluter bound to the run’s `EnvelopeModel`). The three top-down defaults that re-anchor a feature onto the **real averagine** — the multi-envelope refine (`refine_feature_multi`, no model arg), the averagine mono-offset fit (`TD_MONO_FIT`), and IsoDec charge re-assignment — are **disabled**, because on a decoy they would launder the coincidence back

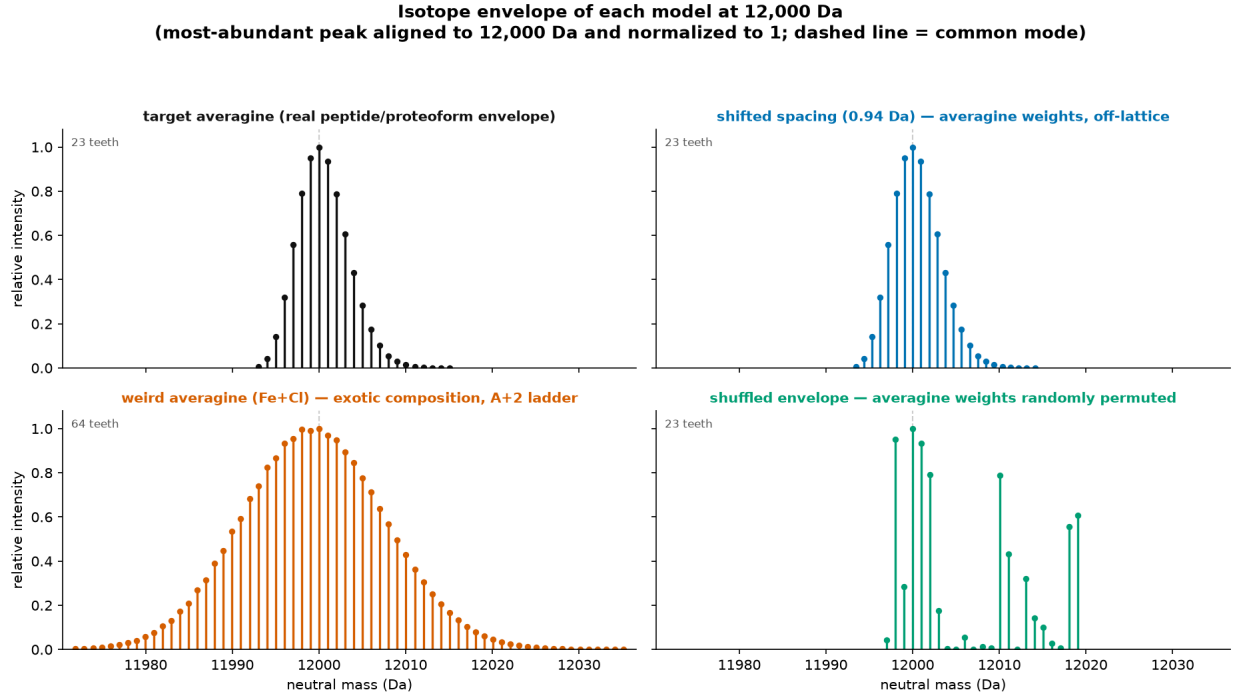


Figure 1: Figure 1 — Isotope envelope of the target averagine and the three decoy models at 12,000 Da. All four are drawn with the most-abundant peak aligned to 12,000 Da (dashed line) and normalized to 1, so only the shape differs: shifted-spacing has the averagine weights on a compressed 0.94-Da lattice; weird (Fe+Cl) is a broad 64-tooth envelope whose A+2 ladder carries the mode far off the mono; shuffled has the averagine weight multiset in a scrambled order.

toward a real pattern and understate separation. The multi-charge detector’s own cross-charge mono-offset refine (MC_MONO_KMAX=3) is left on; it is now decoy-aware.

Scorer (decoy_score_export, SCORE_MAX_ISOTOPES=60 SCORE_RT_SIGMA=0.30) — heavy top-down envelopes span far more than the bottom-up 24-tooth template and elute broader than 0.15 min.

Datasets

key	raw	ID reference (positive class)	RT tol
Golden	D:\CD_FDR_gsvgo000623671.tsvg (930 proteoforms)	MSVgo000623671.tsvg (930 proteoforms)	±1.0 min
Jurkat	D:\JurkatTopdownk02-16-20-segjurkatv (15411 final Classic/IsoDec intersection)	Topdownk02-16-20-segjurkatv (15411 final Classic/IsoDec intersection)	±1.0 min

A target feature is **ID-matched** if its mono is within 20 ppm of a reference proteoform at any $k \cdot (C13-C12)$ offset (k in $[-3, 3]$) and its apex RT within tolerance — isotope-tolerant matching, so a detected-but-off-by-one proteoform still counts as “detected the ID” (the envelope-fit score, not mono precision, is the axis under test). Positive class = ID-matched target; negative class = each decoy.

Feature counts (per-charge refined features)

dataset	target (ID-matched)	shifted	weird	shuffled	detect time / config
Golden	72,571 (6,567 = 9.0%)	76,725	51,672	72,512	54–81 s
Jurkat	885,297 (6,083 = 0.7%)	820,581	556,345	854,500	1300–1657 s

ROC-AUC (ID-matched target vs decoy; 0.5 = none, 1.0 = perfect)

dataset	decoy	AUC 2-D	AUC 3-D
Golden	shifted spacing (0.94 Da)	0.715	0.726
Golden	weird averagine (Fe+Cl)	0.938	0.885
Golden	shuffled envelope	0.896	0.865
Jurkat	shifted spacing (0.94 Da)	0.694	0.669
Jurkat	weird averagine (Fe+Cl)	0.922	0.785
Jurkat	shuffled envelope	0.899	0.795

Golden (930 proteoforms) - ID-matched target vs decoys

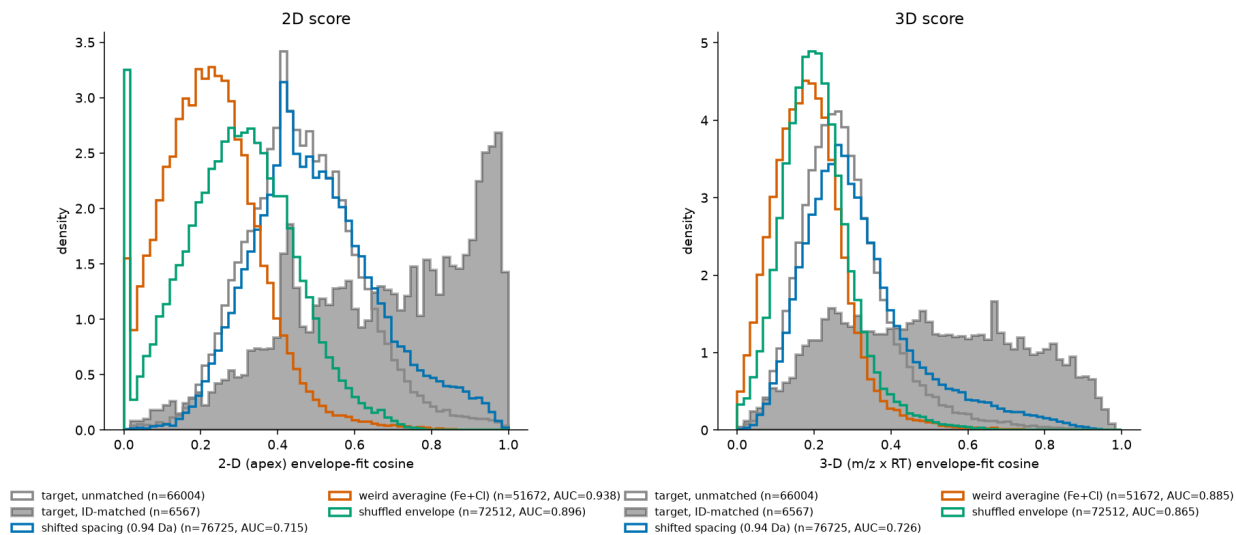
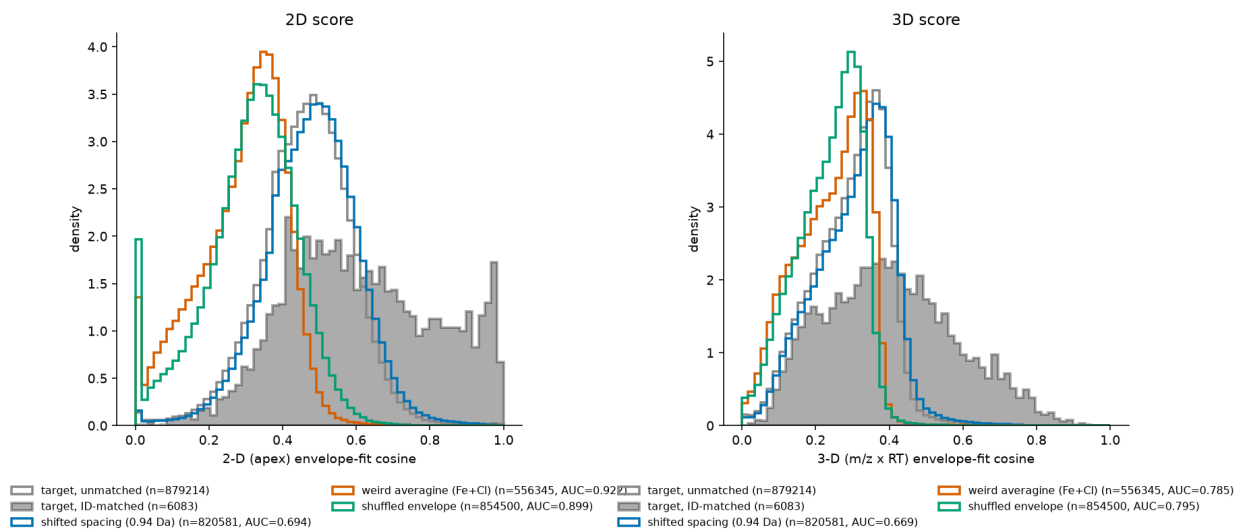


Figure 2: Figure 2 — 2-D and 3-D envelope-fit cosine distributions: unmatched target vs ID-matched target vs the three decoys. Golden (top), Jurkat (bottom). ID-matched real features pile up at high cosine; the weird and shuffled decoys pile up near zero. On heavy Jurkat proteoforms the whole real distribution sits lower (broad envelopes), but the decoys move down with it so separation holds.

Jurkat (1541 hi-conf proteoforms) - ID-matched target vs decoys



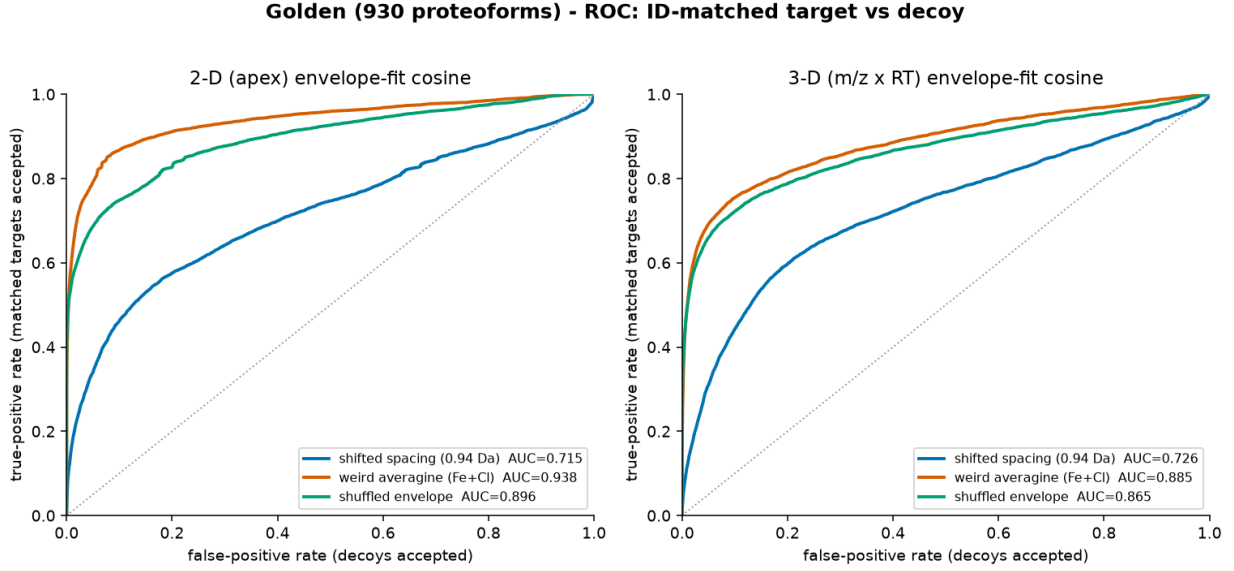
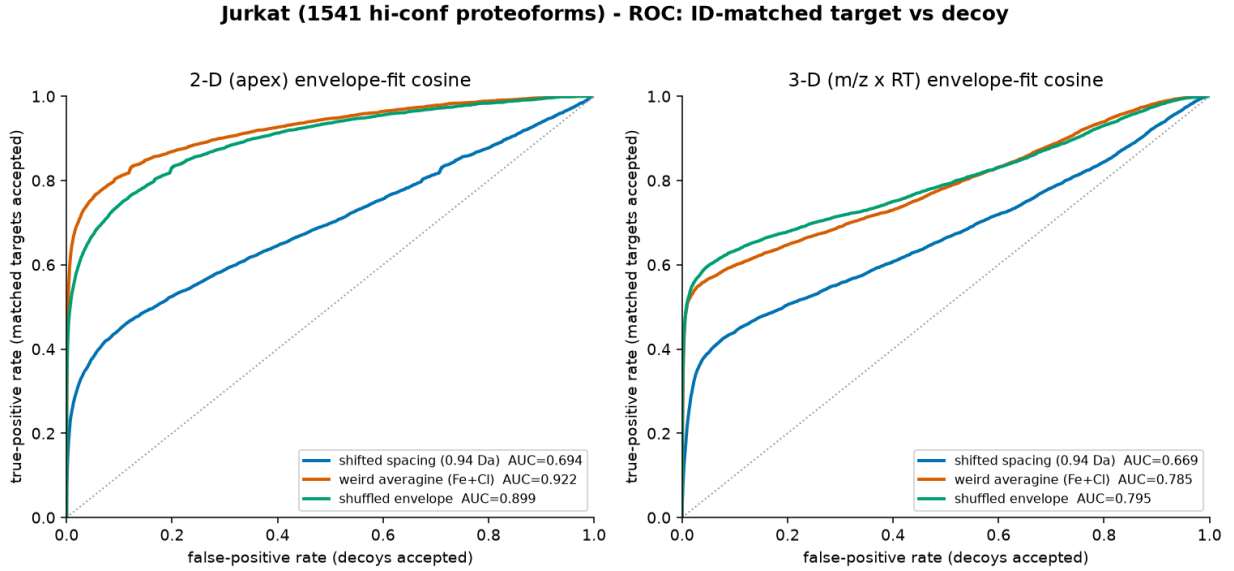


Figure 3: Figure 3 — ROC curves, ID-matched target vs each decoy, 2-D and 3-D score. Golden (top) and Jurkat (bottom): weird averagine (Fe+Cl) and shuffled separate strongly on the 2-D apex score; shifted spacing barely beats chance; the 3-D score is weaker than 2-D on both files.



Operating thresholds (2-D apex cosine)

95% recall threshold — the score at the 5th percentile of ID-matched target scores (retain 95% of real IDs); the decoy pass-rate is the false-positive rate paid for it:

dataset	decoy	threshold	real recall	decoy pass
Golden	shifted	0.248	95%	95.8%
Golden	weird	0.248	95%	41.7%

dataset	decoy	threshold	real recall	decoy pass
Golden	shuffled	0.248	95%	62.8%
Jurkat	shifted	0.305	95%	92.2%
Jurkat	weird	0.305	95%	52.2%
Jurkat	shuffled	0.305	95%	57.4%

95% decoy-rejection threshold — the score at the 95th percentile of that decoy’s scores, i.e. only **5% of decoys pass** (a 5% false-positive / 95% specificity operating point; the correct name for what the bottom-up work loosely called the “precision” threshold). Reports the real-ID recall retained:

dataset	decoy	threshold	real recall	decoy pass
Golden	shifted	0.817	34.0%	5%
Golden	weird	0.441	79.2%	5%
Golden	shuffled	0.538	68.7%	5%
Jurkat	shifted	0.671	38.0%	5%
Jurkat	weird	0.453	75.9%	5%
Jurkat	shuffled	0.500	67.1%	5%

(3-D numbers in `topdown_decoy_summary.json`; 3-D real-recall at 5% decoy pass is ~10–20 pp lower.)

Takeaways

1. **The decoy machinery works end-to-end in top-down.** With decoys threaded into the joint charge-ladder detector, all three separate ID-matched proteoforms from decoys on both datasets — the same result the bottom-up experiment showed, now for the multi-charge path.
2. **Decoy-strength ranking is identical to bottom-up: weird (Fe+Cl) > shuffled > shifted spacing.** The Fe+Cl envelope’s A+2-dominated shape is the hardest for a real proteoform to mimic; at a 5% decoy pass-rate it still retains ~76–79% of real IDs on both files.
3. **Shifted spacing is a weak decoy in top-down — weaker than in bottom-up** (AUC 0.69–0.72 vs 0.79–0.88 bottom-up). A 0.94-Da comb at high charge is only slightly off the 1.0034-Da ¹³C lattice, and top-down spectra are dense and high-charge, so the off-lattice comb still catches real isotope peaks by coincidence. It is close to useless as a top-down decoy.
4. **2-D beats 3-D on both top-down files** (e.g. Jurkat weird 0.922 vs 0.785), like the bottom-up *fast* gradient. The wide-window 3-D integration folds off-apex noise into the norm and dilutes the contrast; the RT axis only helped bottom-up where chromatography was well resolved.
5. **Real-feature cosines sit lower on Jurkat than Golden** (matched mode ~0.45 vs ~0.9). Jurkat proteoforms are heavy (13–21 kDa): broad envelopes, mono well below noise, more unexplained in-window peaks — so even real features score lower. The decoys move down with them, so separation holds, but the absolute thresholds are lower and file-specific (calibrate per file, not with a global cut).

Cross-tool comparison: TopFD features under the same score

As an external check, we scored the features from **TopFD** (an established top-down feature detector) on the same Jurkat raw with the *same* envelope-fit cosine. TopFD’s `*_feature.xml` was parsed (`parse_topfd.py`) into one row per (proteoform, charge) — 20,465 features → 29,014 per-charge rows, `apex_time` converted from seconds to minutes — and scored by `decoy_score_export` against averagine 1.0 with the same settings used for our own features. This asks where an independent tool’s features fall on the axis we built.

population	n	median 2-D score
our weird decoy	556,345	0.311
our shuffled decoy	854,500	0.326
our target, all	885,297	0.470
TopFD, all	29,014	0.534
our target, ID-matched	6,083	0.594

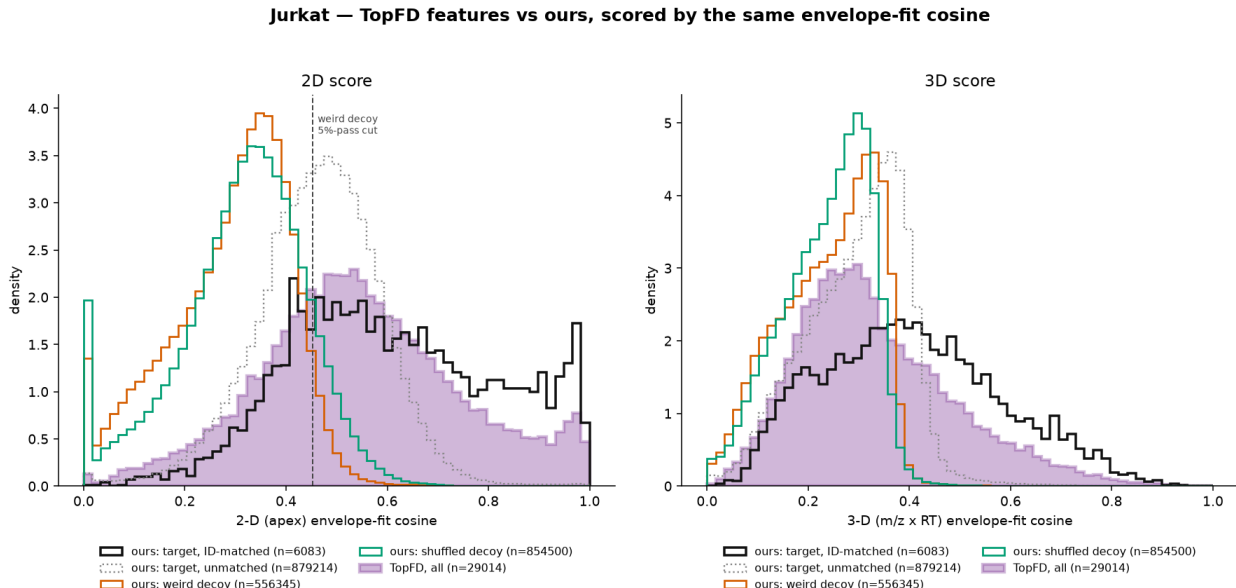


Figure 4: Figure 4 — TopFD’s Jurkat features (filled purple) scored by our envelope-fit cosine, overlaid on our own target (ID-matched and unmatched) and the two strong decoys. TopFD’s features sit well above the decoy floor, between our target-all and our ID-matched distributions. Dashed line = the weird-decoy 5%-pass threshold (2-D = 0.453).

Read. TopFD’s features land clear of the decoy floor and squarely in our real-feature region — a cross-tool sanity check that the score rewards genuine envelope quality, not something idiosyncratic to our detector. Two caveats frame the comparison:

- **TopFD is far more selective.** It reports ~20k proteoform-level features vs our 885k per-charge detections; we emit a large low-intensity tail TopFD never reports, which drags our target-all median (0.470) *below* TopFD’s (0.534). Against the high-confidence GT, TopFD’s ID-match rate is 8.9% vs 0.7% for our per-charge dump — same cause. TopFD’s distribution is best compared to a high-quality subset of ours, not the raw dump.

- **About a third of TopFD's features fall below our decoy-calibrated cut.** At the threshold where only 5% of weird decoys pass ($2\text{-D} = 0.453$), **68.0%** of TopFD features clear it, vs **75.9%** of our ID-matched targets — the shortfall is mostly heavy / high-charge proteoforms and off-by-one monos whose broad envelopes score low (the same effect that lowers our own heavy-Jurkat real features).

Reproduce

```
# detect + score, per dataset (writes t_<ds>_<model>.* and s_<ds>_<model>.tsv)
.\run_topdown_decoys.ps1 golden "D:\CD_FDR_MSV000082367\GoldenRaw_FullTaskBarrage_TD-6-2\Task2
.\run_topdown_decoys.ps1 jurkat "D:\JurkatTopdown\02-18-20_jurkat_td_rep1_fract6.raw"
# distributions + ROC/AUC + thresholds + summary json
python analyze_topdown_decoys.py

# cross-tool: parse TopFD's feature XML, score it the same way, plot vs ours
python parse_topfd.py "D:\JurkatTopdown\MultipleDeconTest\TopFD\02-18-20_jurkat_td_rep1_fract6
$env:SCORE_MAX_ISOTOPES="60"; $env:SCORE_RT_SIGMA="0.30"
& $scr "D:\JurkatTopdown\02-18-20_jurkat_td_rep1_fract6.raw" topfd_jurkat.refined.tsv s_jurkat
python plot_topfd.py
```